

SHIRAZ UNIVERSITY

School of Electrical and Computer Engineering

CSE & IT Department

ASSIGNMENT 4

GRAPH CONVOLUTIONAL NETWORK AND TRANSFORMER

for Molecular Representation Learning and Generation

Bimodal representation alignment and autoregressive SELFIES generation

COURSE

Neural Network & Deep Learning

Technical Report

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Abstract

GRAPH–SEQUENCE ALIGNMENT / MOLECULAR GENERATION

Learning molecular representations that remain consistent across structural graphs and symbolic sequences is important for chemical retrieval and generative modeling. This project presents a joint Graph Convolutional Network (GCN)-Transformer framework that aligns graph-based and SELFIES-based descriptions of molecules while retaining an autoregressive generation capability. Starting from 4,200 compounds in the MoleculeNet Lipophilicity dataset, SMILES strings were parsed, sanitized, canonicalized, converted to SELFIES, and filtered using a 100-token limit, yielding 4,194 usable molecules divided into 3,355 training, 419 validation, and 420 test samples. The graph encoder combines separate embeddings for nine atom features with three residual GCN layers, global mean pooling, and an L2-normalized projection head. The sequence encoder is a four-layer, six-head Transformer with a 192-dimensional hidden space, causal and padding masks, and an independent normalized projection head. Training comprised a 15-epoch language-model warm-up followed by 40 epochs of joint optimization using token-prediction, bidirectional contrastive, and positive-pair losses. On the held-out test set, graph-to-sequence retrieval achieved 50.48% Top-1 and 85.95% Top-5 accuracy, while sequence-to-graph retrieval reached 48.81% and 83.57%, respectively, far above random baselines of 0.238% and 1.19%. Matching pairs had a mean cosine similarity of 0.8806, compared with 0.0306 for nonmatching pairs, indicating effective cross-modal alignment. The Transformer further generated 900 SELFIES sequences at temperatures 0.8, 1.0, and 1.2; all decoded to RDKit-valid molecules. Acceptance rates after the complete validation pipeline were 87.67%, 79.67%, and 64.33%, and every accepted molecule was unique, exactly novel, and structurally novel under a Morgan-fingerprint Tanimoto threshold of 0.85. Mean QED decreased from 0.513 to 0.418 as temperature increased, revealing the expected diversity-quality trade-off. Overall, the results show that GCN and Transformer encoders can learn compatible molecular representations while supporting valid and novel molecular generation. Nevertheless, computational validity and novelty do not establish molecular stability, synthetic accessibility, safety, or biological activity.

Introduction

01 / MULTIMODAL MOLECULAR REPRESENTATION LEARNING

1.1 Background and Motivation

Molecular learning is intrinsically multimodal because a single compound can be expressed as both a topology and a language-like string. In a molecular graph, atoms form nodes and bonds define edges; a Graph Convolutional Network (GCN) aggregates information from bonded neighborhoods and converts variable-size node features into a fixed graph-level embedding through global pooling. The same structure can also be written as a symbolic sequence. SELFIES provides a constrained, tokenized representation that is more robust for generation than raw SMILES, while a Transformer uses self-attention and positional information to model dependencies between tokens and support causal next-token prediction (Kipf and Welling, 2017; Vaswani et al., 2017; Krenn et al., 2020).

The two views emphasize complementary information. The graph encoder is naturally aligned with molecular connectivity, whereas the sequence encoder can learn symbolic regularities and generate structures autoregressively. Training them independently, however, does not guarantee that equivalent molecules occupy compatible regions of representation space. This project therefore treats every graph–SELFIES pair as two observations of the same compound and learns a shared embedding through symmetric contrastive alignment. Bidirectional retrieval tests whether one view can identify its partner in the other, while generation tests whether the sequence model has learned a usable molecular distribution rather than only a static encoding.

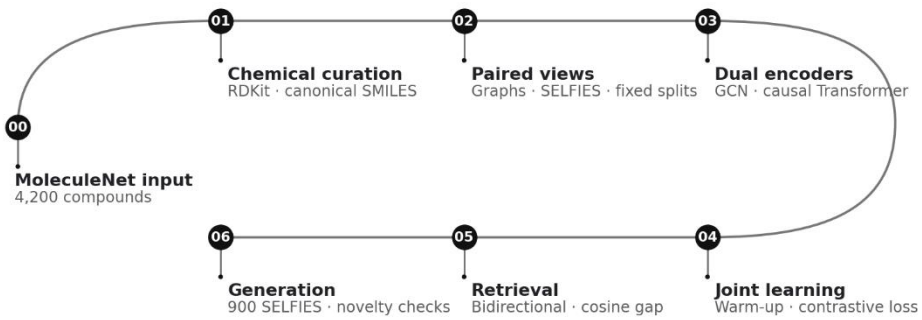
$$\textit{molecule } m \rightarrow \{\textit{ molecular graph } G_m \textit{ , SELFIES sequence } S_m \} \rightarrow \textit{aligned embeddings } z_G \approx z_S$$

1.2 Project Objectives

Using the 4,200-entry MoleculeNet Lipophilicity dataset (Wu et al., 2018), the study pursues three linked objectives:

- **Build reliable paired inputs.** Parse, sanitize, canonicalize, and deduplicate the structures; create fixed 80/10/10 splits; and derive molecular graphs and SELFIES sequences with a training-only vocabulary.
- **Learn a compatible cross-modal space.** Combine a three-layer residual GCN with a four-layer causal Transformer, perform language-model warm-up, and jointly optimize token, bidirectional contrastive, and positive-pair losses.
- **Evaluate representation and generation quality.** Measure retrieval in both directions, generate SELFIES at three temperatures, and apply RDKit, chemical-range, exact-novelty, fingerprint, and nearest-neighbor validation.

The pipeline connects representation learning, generative modeling, and chemical screening within one reproducible experiment. Its final checks establish computational acceptability and novelty only—not molecular stability, synthesizability, safety, or biological utility.



Dataset Preparation and Preprocessing

02 / CHEMICAL CURATION AND PAIRED MOLECULAR INPUTS

2.1 Dataset Description

The experiment used the Lipophilicity subset of MoleculeNet, loaded through the PyTorch Geometric MoleculeNet interface. The raw collection contains 4,200 compounds. Each record provides a molecular structure as a SMILES string, categorical atom and bond information, molecular connectivity, and an experimental lipophilicity target. The target was retained in each graph object as y, while the representation-learning stages described in this report were optimized with sequence-modeling and graph–sequence alignment losses.

Every accepted compound was converted into two synchronized views: a molecular graph for the GCN and a SELFIES sequence for the Transformer. The original record index, canonical SMILES, SELFIES string, target value, atom count, and undirected bond count were stored together so the two encoders always received representations of the same molecule.

RAW SMILES → RDKIT SANITIZE → **CANONICAL ISOMERIC SMILES** → SELFIES → **PAIRED RECORD**

2.2 SMILES Parsing and Canonicalization

Each original SMILES was parsed with RDKit using sanitization, followed by an explicit Chem.SanitizeMol call. The sanitized molecule was converted to canonical isomeric SMILES with canonical=True and isomericSmiles=True, then reparsed as a final structural check. Canonicalization was performed before splitting because different SMILES strings can encode the same molecule; uniqueness was therefore enforced on molecular identity rather than raw text.

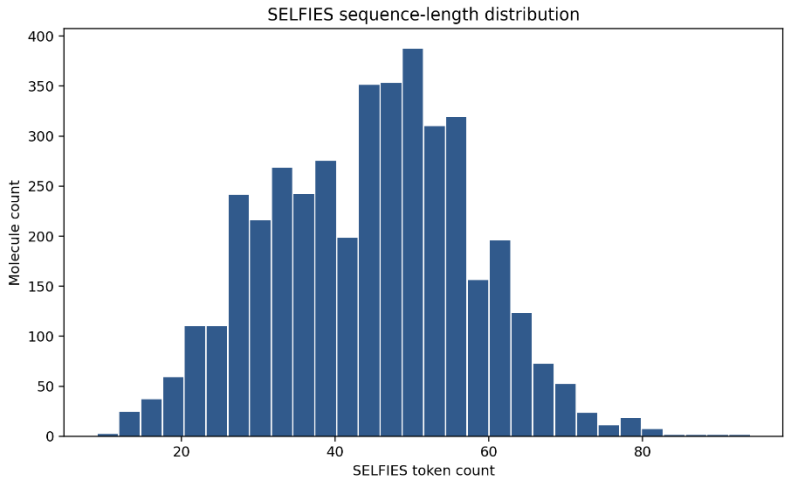
A set of previously observed canonical strings was used for deterministic duplicate removal. All 4,200 original structures parsed and sanitized successfully, and no canonical duplicate was found. Canonical molecules were then encoded as SELFIES. Empty conversions, conversion failures, and sequences longer than 100 SELFIES tokens were rejected. Six overlength sequences were removed, leaving 4,194 usable graph–sequence pairs.

For graph preprocessing, each canonical SMILES was converted with torch_geometric.utils.from_smiles. This produced nine integer-coded categorical atom fields. Every undirected chemical bond was expanded into two directed edges, and its approximate bond order was stored as a scalar edge weight; the weighting scheme is detailed in Section 3.2.

CURATION RESULT **4,200 entered · 4,200 RDKit-valid · 0 canonical duplicates · 6 overlength sequences removed · 4,194 usable pairs**

2.3 SELFIES Tokenization

SELFIES strings were segmented with split_selfies so each grammar symbol became one discrete token. The vocabulary was fitted only on the training split and combined with four control tokens: <pad>, <bos>, <eos>, and <unk>. This produced 52 token IDs. Each sequence was wrapped with <bos> and <eos> and padded to 102 positions (100 content tokens plus two boundary tokens). The validation set contained no unknown tokens; the test set contained only two occurrences.



2.4 Dataset Splitting

After global canonical deduplication and length filtering, a fixed random permutation was generated with torch.Generator(seed=42). The first 80% formed the training set, the next 10% the validation set, and the remainder the test set, yielding 3,355, 419, and 420 molecules. The indices were saved for exact reproducibility. Building the vocabulary after this split prevents validation and test tokens from influencing the learned token inventory.

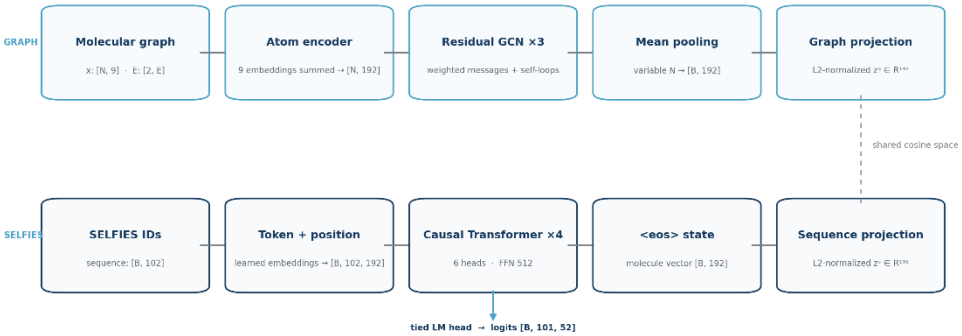
2.5 Dataset Statistics

Metric	Observed value	Definition
Original molecules	**4,200**	Raw MoleculeNet Lipophilicity records
Invalid structures	**0**	Failed RDKit parsing or sanitization
Canonical duplicates	**0**	Repeated canonical isomeric SMILES
SELFIES conversion failures	**0**	Exceptions or empty SELFIES strings
Overlength SELFIES	**6**	More than 100 content tokens
Usable molecules	**4,194**	Final paired graph–sequence records
Train / validation / test	**3,355 / 419 / 420**	Fixed 80% / 10% / 10% split
Vocabulary size	**52**	48 training symbols + 4 special tokens
Average atom count	**26.97**	Atoms per accepted molecule
Average undirected bonds	**29.42**	Chemical bonds before edge duplication
SELFIES length	**9 / 44.39 / 94**	Minimum / mean / maximum token count
Unknown tokens	**0 / 2**	Validation / test occurrences

Proposed Model Architecture

03 / DUAL-ENCODER MOLECULAR REPRESENTATION MODEL

The proposed system is a paired molecular representation model with two independent encoders. A residual graph convolutional network summarizes molecular topology, while a causal Transformer models the corresponding SELFIES sequence. Both branches produce 192-dimensional molecule-level vectors and use separate projection networks to enter a shared, L2-normalized alignment space. The Transformer additionally retains a tied language-model head for next-token prediction and later autoregressive generation.



PAIRED VIEWS · INDEPENDENT ENCODERS · SHARED 192-D COSINE SPACE

3.1 GCN Molecular Encoder

Each atom is described by nine integer-coded categorical fields from the PyTorch Geometric SMILES schema. Their cardinalities are 119, 9, 11, 12, 9, 5, 8, 2, and 2. Rather than treating these IDs as continuous quantities, the encoder learns one 192-dimensional embedding table per field and sums the nine vectors for each atom. This creates an initial node matrix $H^{(0)} \in R^{N \times 192}$ while preserving the semantic separation between atom attributes.

Three residual GCN blocks then exchange messages along molecular bonds. Each block uses GCNConv with normalized aggregation and automatically added self-loops, followed in the implemented order by GELU, LayerNorm, dropout (p = 0.10), and a residual addition. Stacking three blocks gives every atom access to information up to three graph hops away, consistent with the message-passing interpretation in the course GNN notes.

After a final LayerNorm, global mean pooling averages all node embeddings belonging to the same molecule. This operation converts variable-size graphs into a fixed $G \in R^{B \times 192}$ representation without depending on atom ordering. A two-layer projection network ($192 \rightarrow 192 \rightarrow 192$, GELU and dropout) then produces the unit-normalized graph vector z^g used for contrastive alignment.

$$\mathbf{h}_v^{(0)} = \sum_{f=1}^{9} \mathbf{E}_f(x_{v,f})$$

$$\mathbf{M}^{(\ell)} = \hat{\mathbf{D}}^{-\frac{1}{2}} \hat{\mathbf{A}}_w \hat{\mathbf{D}}^{-\frac{1}{2}} \mathbf{H}^{(\ell)} \mathbf{W}^{(\ell)}$$

$$\mathbf{H}^{(\ell+1)} = \mathbf{H}^{(\ell)} + \mathrm{Dropout}(\mathrm{LN}(\mathrm{GELU}(\mathbf{M}^{(\ell)})))$$

$$\mathbf{g}_b = \frac{1}{|\mathcal{V}_b|} \sum_{v \in \mathcal{V}_b} \mathrm{LN}(\mathbf{h}_v^{(3)})$$

3.2 Bond-Order Edge Weighting

The graph connectivity supplied to GCNConv is directed: every undirected chemical bond {u, v} is expanded into (u, v) and (v, u), and both directions receive the same scalar weight. The adjacency therefore encodes topology, while the scalar edge_weight tensor modulates normalized message strength according to approximate bond order.

Single, aromatic, double, and triple bonds are assigned 1.0, 1.5, 2.0, and 3.0, respectively; any unrecognized type falls back to 1.0. The mapping supplies more chemical information than an unweighted adjacency matrix, but it remains a deliberate simplification: it is not a learned bond embedding and does not fully represent stereochemistry, conjugation, ring membership, or other categorical edge attributes.

SINGLE	AROMATIC	DOUBLE	TRIPLE
1.0	1.5	2.0	3.0

3.3 Transformer Sequence Encoder

The sequence branch is a four-layer TransformerEncoder used in causal language-model mode. SELFIES token IDs are mapped through a 52 × 192 embedding matrix; learned positional embeddings cover all 102 padded positions. Token vectors are scaled by √192, added to the positional vectors, and passed through embedding dropout before attention.

Each pre-normalized layer contains six attention heads (32 dimensions per head), a 512-dimensional position-wise feed-forward network, GELU activation, residual connections, and dropout of 0.10. The causal mask converts the otherwise bidirectional encoder stack into an autoregressive sequence model. The hidden state at <eos> is selected as the molecule-level sequence embedding; if <eos> is absent, the last non-padding state is used. A separate 192 → 192 → 192 projection network yields unit-normalized z^s. The language-model output matrix is weight-tied to the token embedding matrix.

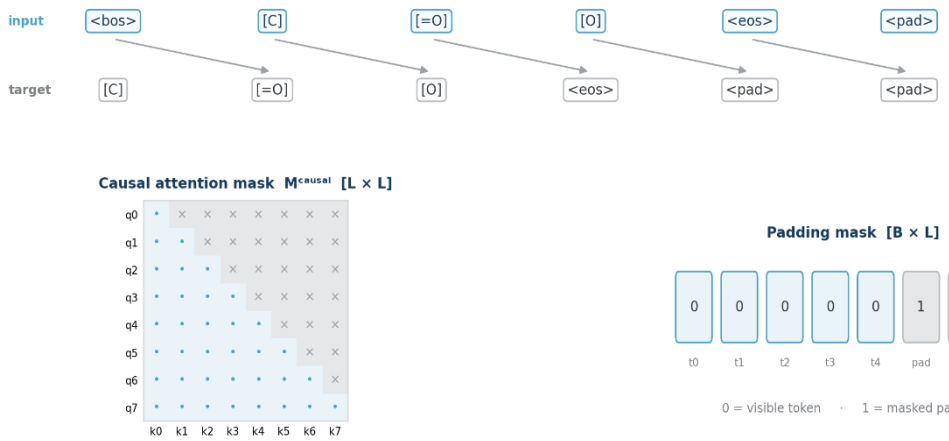
$$\text{Attention}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\top}}{\sqrt{d_k}} + \mathbf{M}\right)\mathbf{V}, \qquad d_k = 192/6 = 32$$

Component	Implemented configuration
Total / trainable parameters	**1,710,080 / 1,710,080**
Atom embedding tables	**9 separate categorical tables**
GCN encoder	**3 residual blocks; 192 hidden units**
Bond weights	**1.0 / 1.5 / 2.0 / 3.0**
Transformer encoder	**4 pre-norm causal layers**
Attention	**6 heads; 32 dimensions per head**
Feed-forward network	**512 dimensions; GELU**
Token / position length	**52-token vocabulary / 102 positions**
Projection networks	**192 → 192 → 192; L2 normalization**
Regularization	**Dropout = 0.10**

3.4 Shifted Input and Target

Teacher forcing is implemented by a one-position shift. For a complete padded sequence S = [<bos>, s_1, …, s_n, <eos>, <pad>, …], hidden state h_t is trained to predict S_{t+1}. Operationally, logits are computed from hidden[:, :-1, :] and labels are taken from sequence_ids[:, 1:]. With a padded length of 102, the model therefore returns 101 prediction positions and 52 vocabulary logits per position: [B, 101, 52].

Cross-entropy is evaluated only at non-padding labels through ignore_index = PAD_ID. The shift gives every supervised position a next-token target, including prediction of <eos>, while the causal mask ensures that the correct future token cannot be copied from the input during training.



3.5 Causal and Padding Masks

Two Boolean masks serve different purposes. The L × L causal mask marks every location above the main diagonal, $M^{causal}_{ij} = 1$ when j > i, preventing position i from attending to later tokens. The B × L padding mask is sequence_ids.eq(PAD_ID), preventing padded keys from contributing to attention. The loss mask is applied separately by ignoring padded target IDs in cross-entropy.

Together, these constraints preserve parallel computation across positions while enforcing the same information boundary used at generation time: a token may depend on itself and previous tokens, never on future or padded content. This directly follows the masked self-attention and one-position offset described in the course Transformer notes.

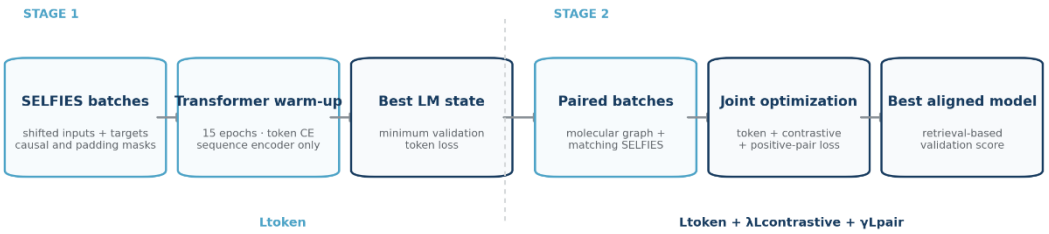
INFORMATION BOUNDARY **future keys are masked · padded keys are masked · padded labels are excluded from cross-entropy**

Training Strategy

04 / TWO-STAGE AUTOREGRESSIVE AND CONTRASTIVE OPTIMIZATION

Optimization was organized as a two-stage curriculum. The causal Transformer was first trained as a standalone SELFIES language model so that token embeddings, positional embeddings, masked attention, and the tied output head acquired a stable autoregressive solution. The complete graph–sequence model was then

fine-tuned jointly, preserving next-token prediction while aligning molecular graph and SELFIES representations in a shared cosine space.



4.1 Language-Model Warm-up

During the first 15 epochs, only the parameters of the sequence encoder were supplied to the optimizer; the GCN and both projection networks therefore received no updates. Each padded sequence was trained with teacher forcing and a one-position offset. Cross-entropy was accumulated over non-padding targets only, so <pad> positions affected neither the loss nor token accuracy.

AdamW used an initial learning rate of 3×10^{-4} and weight decay of 10^{-3}. A cosine annealing schedule reduced the rate to 5% of its initial value over the 15 epochs. Automatic mixed precision and a maximum gradient norm of 1.0 were applied at every training step. The sequence-encoder state with the lowest validation token loss was restored and used to initialize joint training.

$$\mathcal{L}_{\text{token}} = -\frac{1}{N_{\text{valid}}} \sum_{\{(b,t)\,:\,y_{b,t} \neq \text{pad}\}} \log p_\theta(y_{b,t} \mid y_{b,<t})$$

4.2 Joint GCN–Transformer Training

The second stage activated all 1,710,080 trainable parameters for up to 40 additional epochs. To protect the pretrained language model while allowing the new alignment path to adapt quickly, AdamW used two parameter groups: 5×10^{-5} for the Transformer and 3×10^{-4} for the GCN plus the graph and sequence projection networks. Both groups followed the same cosine multiplier from 1.00 to 0.05.

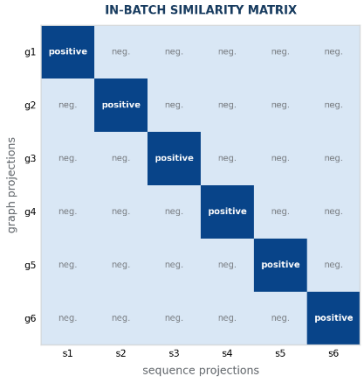
Every mini-batch jointly evaluated token prediction, symmetric graph–sequence retrieval, and direct positive-pair similarity. Validation retrieval was computed after each epoch. Training could stop only after epoch 15 and after eight consecutive epochs without a selection-score improvement greater than 10^{-5}; otherwise it continued to epoch 40. The checkpoint with the highest retrieval-based selection score was retained.

OPTIMIZATION ORDER causal language-model warm-up · joint graph–sequence alignment · retrieval-selected checkpoint

4.3 Contrastive Alignment

The graph and sequence projection networks output L2-normalized vectors z^g and z^s in R^{192}. For a mini-batch of B paired molecules, their dot products form a B × B cosine similarity matrix. Entry (i, i) is the correct graph–SELFIES pair, while the B − 1 off-diagonal entries in the same row and column act as in-batch negatives. With the configured batch size of 256, a full batch therefore supplies 255 negatives per anchor.

Similarities were divided by $\tau = 0.07$ before cross-entropy. The graph-to-sequence term classifies the correct sequence for each graph, and the sequence-to-graph term applies the same criterion to the transposed matrix. Their mean defines the symmetric contrastive loss. A separate positive-pair loss, $1 - \cos(z^g_i, z^s_i)$, directly pulls matched embeddings together even when retrieval ranking is already correct.



$$s_{ij} = \frac{(\mathbf{z}_i^g)^\top \mathbf{z}_j^s}{\tau}, \qquad \tau = 0.07$$

$$\mathcal{L}_{g \to s} = \mathrm{CE}(\mathbf{S}, \mathbf{y}), \qquad \mathcal{L}_{s \to g} = \mathrm{CE}(\mathbf{S}^\top, \mathbf{y})$$

$$\mathcal{L}_{\text{contrastive}} = \frac{1}{2}(\mathcal{L}_{g \to s} + \mathcal{L}_{s \to g})$$

$$\mathcal{L}_{\text{pair}} = \frac{1}{B}\sum_{i=1}^{B}[1 - (\mathbf{z}_i^g)^\top \mathbf{z}_i^s]$$

$\mathbf{y} = [0, 1, \ldots, B-1]$ · projected vectors are L2-normalized

4.4 Combined Objective

The joint objective retains chemical-language competence while learning cross-modal identity. The token term supplies local autoregressive supervision, the contrastive term separates each molecule from all nonmatching batch items in both retrieval directions, and the pair term raises the absolute cosine similarity of corresponding views. The implemented weights were λ = 1.00 and γ = 0.25.

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{token}} + \lambda \mathcal{L}_{\text{contrastive}} + \gamma \mathcal{L}_{\text{pair}}, \qquad \lambda = 1.00, \quad \gamma = 0.25$$

4.5 Training Configuration

Training element	Implemented configuration
Random seed	**42**
Batch size	**256 graph–sequence pairs**
Warm-up horizon	**15 epochs; Transformer only**
Joint horizon	**Up to 40 epochs; minimum 15**
Optimizer / decay	**AdamW / 1×10^{-3}**
Warm-up learning rate	**3×10^{-4}**
Joint learning rates	**Transformer 5×10^{-5}; alignment 3×10^{-4}**
Learning-rate schedule	**Cosine decay to 5% of initial rate**
Numerical stabilization	**Mixed precision; gradient clip = 1.0**
Contrastive settings	**τ = 0.07; λ = 1.00; γ = 0.25**
Early stopping	**Patience 8; minimum $\Delta = 1 \times 10^{-5}$**
Checkpoint selection	**mean bidirectional Top-5 + 0.10 × max(cosine gap, 0)**

Training Results

05 / OPTIMIZATION CONVERGENCE AND REPRESENTATION ALIGNMENT

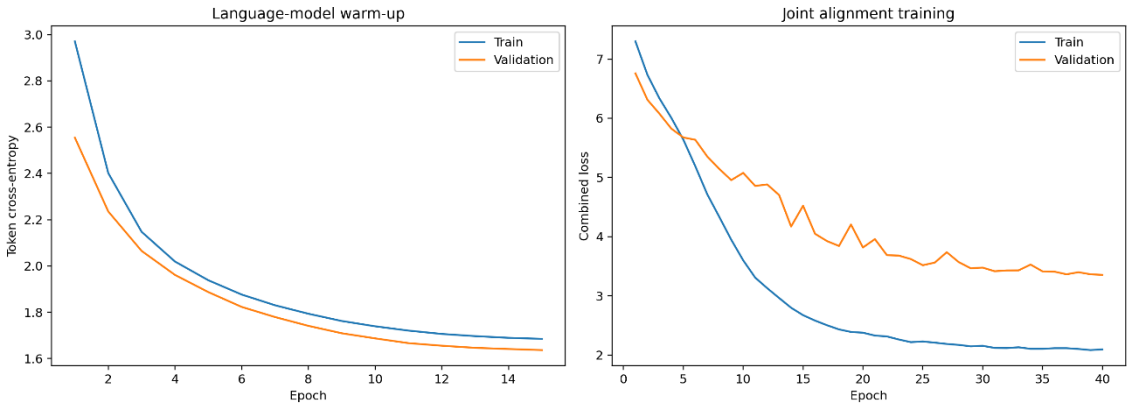
Both optimization stages converged without numerical instability. The language-model warm-up established a usable SELFIES predictor before graph–sequence coupling, and joint training then reduced the alignment-dominated objective while preserving token modeling. All curves below are taken directly from the completed run: 15 warm-up epochs followed by 40 joint epochs.

15	40	37	79.38 s
WARM-UP EPOCHS	JOINT EPOCHS	SELECTED EPOCH	MODEL TRAINING

5.1 Warm-up and Joint Loss

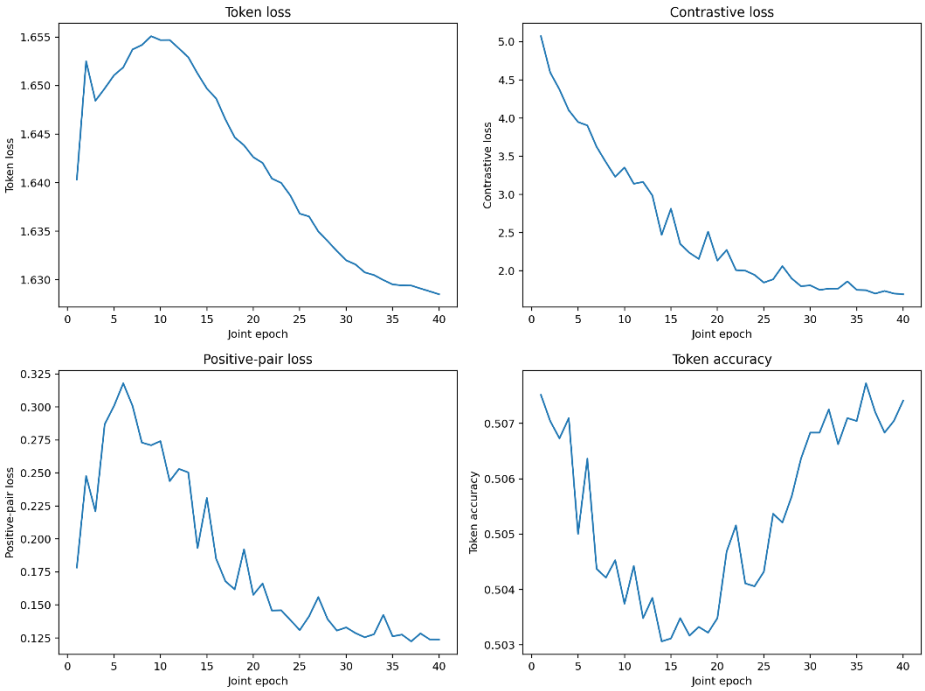
Warm-up cross-entropy decreased smoothly throughout all 15 epochs. Training loss fell from 2.9707 to 1.6849, while validation loss fell from 2.5546 to its minimum of 1.6363. Over the same interval, validation token accuracy increased from 35.43% to 50.98% and validation perplexity decreased from 12.8657 to 5.1363. The closely tracking curves and absence of a late validation reversal indicate stable language-model convergence.

At the start of joint optimization, the combined objective was dominated by cross-modal alignment. Training total loss decreased from 7.2981 to 2.0944 (71.30%), and validation total loss decreased from 6.7578 to 3.3531 (50.38%). Validation improved more gradually and with small oscillations because retrieval is evaluated across the complete validation set, whereas optimization uses changing in-batch negatives.



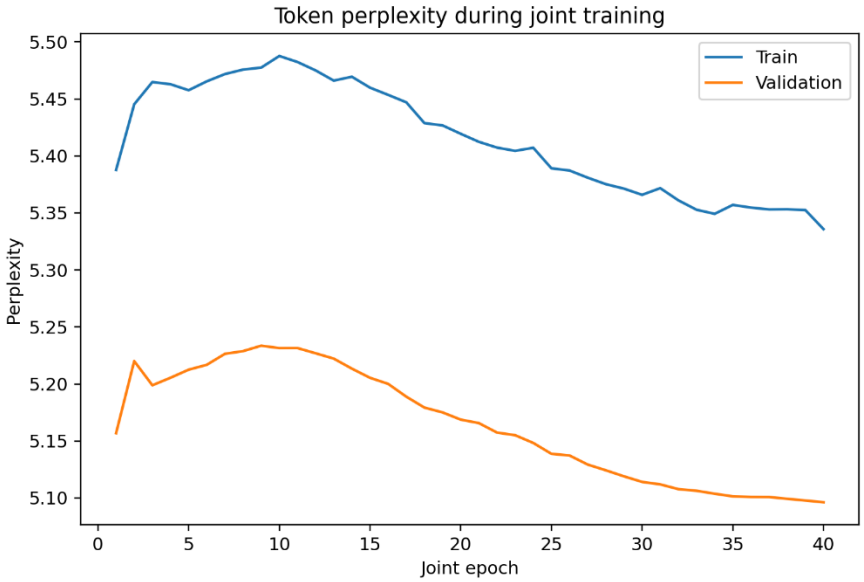
5.2 Joint Loss Components

The validation contrastive term produced the largest change, declining from 5.0729 to 1.6936 (66.61%). Positive-pair loss reached 0.1224 at the selected checkpoint, equivalent to an average matching-pair cosine of 0.8776. In contrast, token loss remained within a narrow range and ended at 1.6285, while token accuracy remained near 50.7%. This separation shows that most joint-stage improvement came from representation alignment rather than from sacrificing the pretrained autoregressive objective.



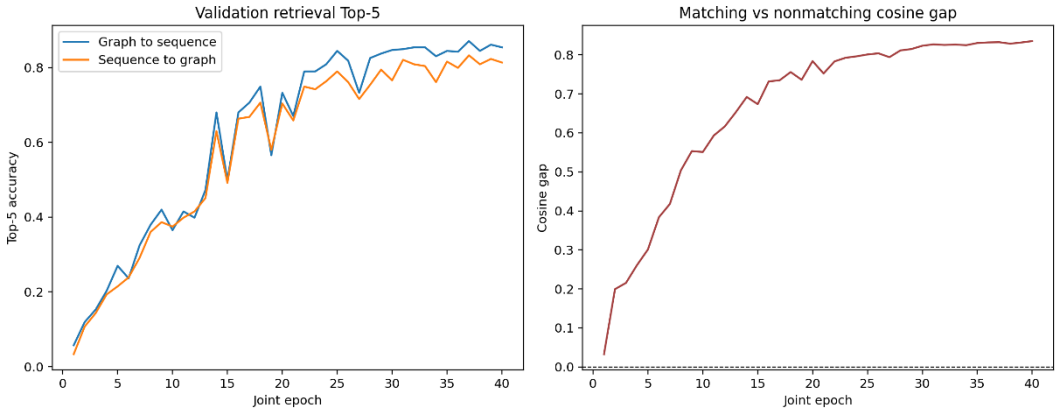
5.3 Perplexity

Token perplexity stayed stable during multimodal fine-tuning. Validation perplexity changed only from 5.1568 at joint epoch 1 to 5.0962 at epoch 40, and the selected epoch-37 model recorded 5.1008. The validation curve remained below the training curve because dropout and training-time regularization are active only on the optimization batches. The important result is preservation of sequence-model quality while alignment improved substantially.



5.4 Alignment Diagnostics

Validation retrieval and cosine separation improved together. From epoch 1 to the selected epoch 37, graph-to-sequence Top-5 increased from 5.73% to 87.11%, sequence-to-graph Top-5 increased from 3.34% to 83.29%, and the matching–nonmatching cosine gap expanded from 0.0327 to 0.8325. These parallel trends provide direct evidence that the two encoders learned a compatible shared representation.



The highest retrieval-based selection score, 0.9353, occurred at epoch 37 and that checkpoint was retained. Validation total loss continued to improve slightly through epoch 40 (3.3531 versus 3.3628 at epoch 37), and the cosine gap reached 0.8353, but both Top-5 retrieval directions declined. Model selection therefore behaved as intended: it favored cross-modal identification performance rather than the smallest aggregate loss alone.

Validation metric	Epoch 1	Epoch 37*	Epoch 40
Validation total loss	6.7578	**3.3628**	3.3531
Validation token loss	1.6403	**1.6294**	1.6285
Validation perplexity	5.1568	**5.1008**	5.0962
Graph→sequence Top-5	5.73%	**87.11%**	85.44%
Sequence→graph Top-5	3.34%	**83.29%**	81.38%
Matching cosine	0.8216	**0.8776**	0.8762
Nonmatching cosine	0.7889	**0.0451**	0.0409
Cosine gap	0.0327	**0.8325**	0.8353
Selection score	0.0486	**0.9353**	0.9177

Bidirectional Retrieval Results

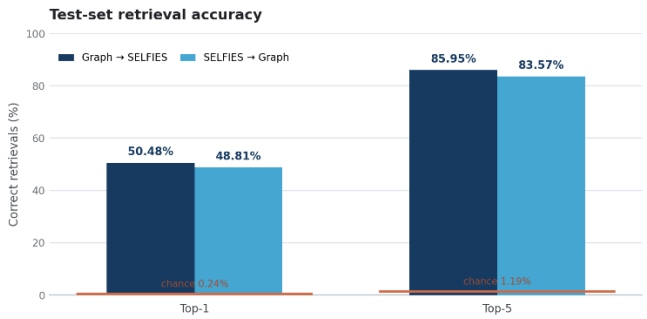
06 / CROSS-MODAL TEST-SET IDENTIFICATION

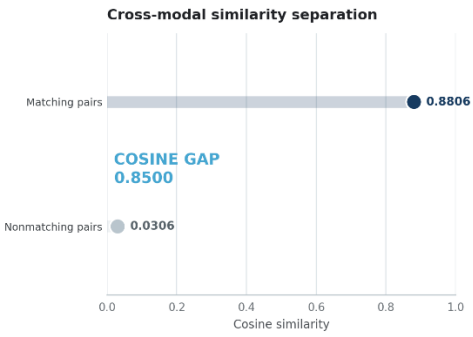
The selected epoch-37 checkpoint was evaluated on the fixed test split by exhaustive all-to-all retrieval. Each of the 420 graph embeddings was compared with every SELFIES embedding using cosine similarity, and the complete ranking procedure was then repeated in the reverse direction. No test-time candidate subsampling or re-ranking was applied.

50.48%	85.95%	48.81%	83.57%
GRAPH → SELFIES TOP-1	GRAPH → SELFIES TOP-5	SELFIES → GRAPH TOP-1	SELFIES → GRAPH TOP-5

RETRIEVAL PROTOCOL

Because both projected embeddings are L2-normalized, their dot product is cosine similarity. For query i, retrieval is counted as correct when the paired representation appears among the K highest-scoring candidates: Top-K = (1/N) Σ_i 1[rank(i) ≤ K], with N = 420.





Retrieval was strong and nearly symmetric. Graph→SELFIES identified 212 exact partners at rank 1 and 361 within the first five positions; SELFIES→Graph recovered 205 and 351, respectively. The small directional differences—1.67 percentage points at Top-1 and 2.38 points at Top-5—suggest balanced cross-modal alignment rather than a representation that works well in only one direction.

Table 5. Exhaustive bidirectional retrieval on the fixed test split

Metric	Graph → SELFIES	SELFIES → Graph	Chance	Fold over chance*
Top-1	50.48% (212/420)	48.81% (205/420)	0.24%	212× / 205×
Top-5	85.95% (361/420)	83.57% (351/420)	1.19%	72.2× / 70.2×

Matching graph–SELFIES pairs reached an average cosine similarity of 0.8806, whereas the average across all nonmatching pairs was only 0.0306, producing a separation gap of 0.8500. Relative to uniform random ranking, Top-1 performance was 212× and 205× higher than chance, and Top-5 performance was 72.2× and 70.2× higher. Together, the retrieval and cosine results show that molecule identity is encoded consistently across the two modalities.

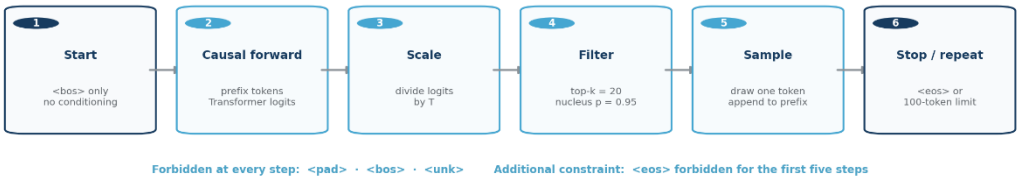
Molecular Generation

07 / UNCONDITIONAL AUTOREGRESSIVE SELFIES SAMPLING

The language-model branch of the selected epoch-37 checkpoint was used as an unconditional molecular generator. Sampling was performed directly in SELFIES token space, where every sequence began from the same <bos> symbol and was extended one token at a time. No molecular graph, property label, test molecule, or ground-truth continuation was supplied at generation time.

900	3	20	0.95
TOTAL SEQUENCES	TEMPERATURES	TOP-K	NUCLEUS TOP-P

7.1 Generation Procedure



At each step, the Transformer received only the tokens already sampled. The current-step logits were divided by the selected temperature and converted to probabilities. Candidate support was then restricted by top-k filtering with k = 20 and nucleus filtering with p = 0.95 before one token was drawn from the resulting categorical distribution.

$$p_T(v \mid y_{<t}) = \mathrm{softmax}(\ell_t(v)/T), \qquad v \in \mathcal{C}_t = \mathrm{TopK}_{20} \cap \mathrm{Nucleus}_{0.95}$$

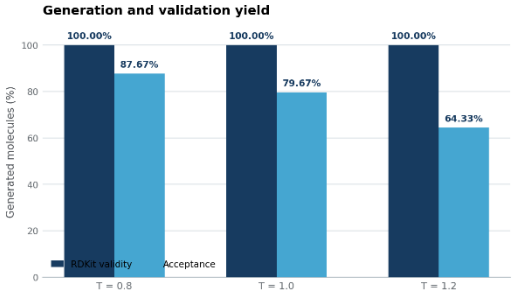
One token is sampled from the filtered distribution and appended to the causal prefix.

The special tokens <pad>, <bos>, and <unk> were assigned zero sampling probability. The <eos> token was also suppressed during the first five generation steps, preventing empty or trivially short outputs. Sampling stopped when <eos> appeared or when the configured maximum of 100 SELFIES tokens was reached. Three independent batches of 300 sequences were generated at T = 0.8, 1.0, and 1.2.

7.2 Generation Results

All 900 sampling trajectories produced <eos> before the maximum-length boundary, and every decoded output passed RDKit parsing and sanitization. After the complete chemistry-aware validation pipeline, 695 molecules remained accepted: 263 at T = 0.8, 239 at T = 1.0, and 193 at T = 1.2. Every accepted canonical structure was unique, exactly absent from the training set, and below the configured structural-similarity threshold.

100%	100%	695	77.22%
EOS TERMINATION	RDKIT VALIDITY	ACCEPTED MOLECULES	OVERALL ACCEPTANCE



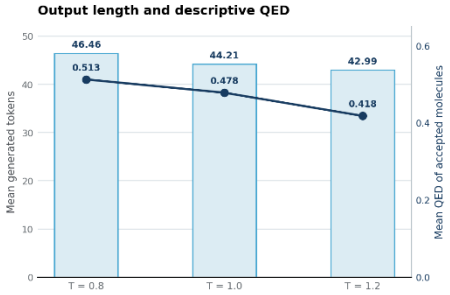


Table 6. Molecular-generation results at each sampling temperature

T	Mean tokens	RDKit validity	Acceptance	Uniqueness	Exact novelty	Structural novelty	Mean QED*
0.8	46.46 [12–99]	100.00% (300/300)	**87.67% (263/300)**	100.00% (263)	100.00% (263)	100.00% (263)	0.5129
1.0	44.21 [9–96]	100.00% (300/300)	**79.67% (239/300)**	100.00% (239)	100.00% (239)	100.00% (239)	0.4784
1.2	42.99 [6–98]	100.00% (300/300)	**64.33% (193/300)**	100.00% (193)	100.00% (193)	100.00% (193)	0.4183

Increasing temperature from 0.8 to 1.2 reduced acceptance from 87.67% to 64.33%, a decrease of 23.34 percentage points. Mean sequence length also declined from 46.46 to 42.99 tokens, while mean QED decreased from 0.5129 to 0.4183. Thus, the tested temperatures preserved perfect canonical uniqueness and novelty, but the higher-temperature samples more often failed downstream chemical-distribution constraints. QED is reported only as a descriptive property and is not interpreted as evidence of molecular usefulness.

Generated-Molecule Validation

08 / STAGED CHEMISTRY-AWARE SCREENING

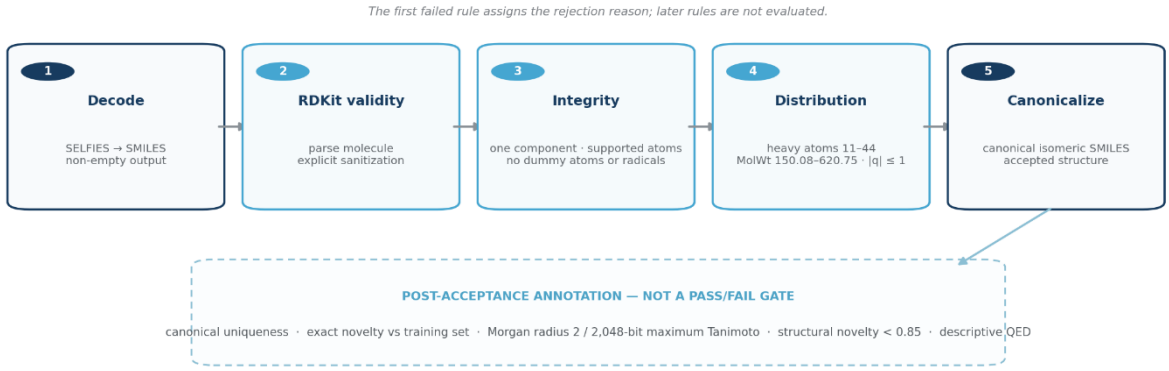
Generated SELFIES were not treated as successful molecules merely because they could be decoded. Each of the 900 samples was processed by a deterministic, first-failure validation pipeline that separated representation validity, RDKit chemical validity, molecular-integrity checks, and training-distribution constraints. This produced 695 accepted structures and 205 rejected structures, corresponding to an overall acceptance rate of 77.22%.

900	695	205	77.22%
ASSESSED	ACCEPTED	REJECTED	ACCEPTANCE

8.1 Complete Validation Pipeline

Validation began with SELFIES decoding and a non-empty SMILES check, followed by RDKit parsing without sanitization and then explicit sanitization. RDKit-valid molecules were required to contain one connected component, no dummy atoms, only elements observed in the training split, and no radical electrons. Heavy-atom count, molecular weight, and absolute formal charge were then checked against training-derived limits before canonical isomeric SMILES were produced.

Figure 9. Complete first-failure validation pipeline and post-acceptance annotations.



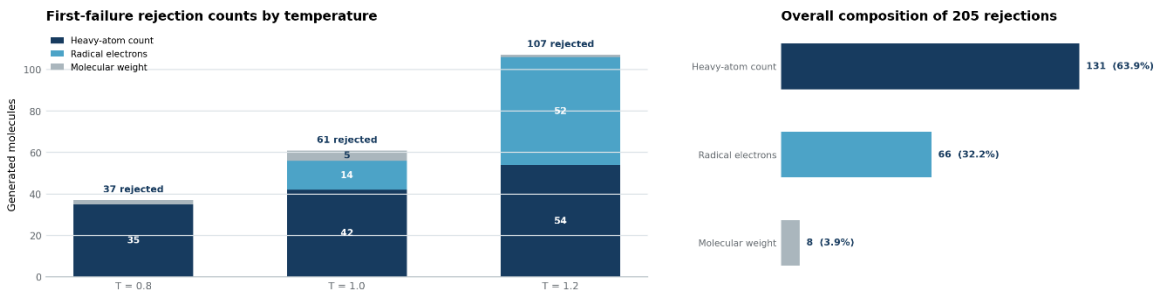
Canonical uniqueness, exact novelty relative to the training canonical-SMILES set, maximum Morgan-fingerprint similarity, structural novelty, and QED were computed after acceptance. They therefore describe the accepted set but did not cause a molecule to pass or fail. Structural novelty used a strict maximum Tanimoto similarity threshold below 0.85 with radius-2, 2,048-bit Morgan fingerprints.

Check	Operational rule	Role
Representation and RDKit	Non-empty decode; parse + sanitize	**Acceptance gate**
Molecular integrity	1 component; no dummy/radical; elements: B, C, N, O, F, Si, P, S, Cl, Se, Br, I	**Acceptance gate**
Heavy-atom count	11–44	**Training P1–P99**
Molecular weight	150.08–620.75 Da	**Training P1–P99**
Absolute formal charge	≤ 1	**Training maximum**
Structural novelty	Exact novel and maximum Morgan Tanimoto < 0.85	**Post-acceptance label**
QED	Computed without thresholding	**Descriptive only**

8.2 Rejection Counts

Only three rejection reasons were observed. Heavy-atom count outside the 1st–99th percentile interval accounted for 131 of 205 rejections (63.90%), nonzero radical electrons accounted for 66 (32.20%), and molecular weight outside the training interval accounted for 8 (3.90%). All decoding, parsing, sanitization, connectivity, dummy-atom, unseen-element, charge, and empty-canonical-SMILES rejection counts were zero.

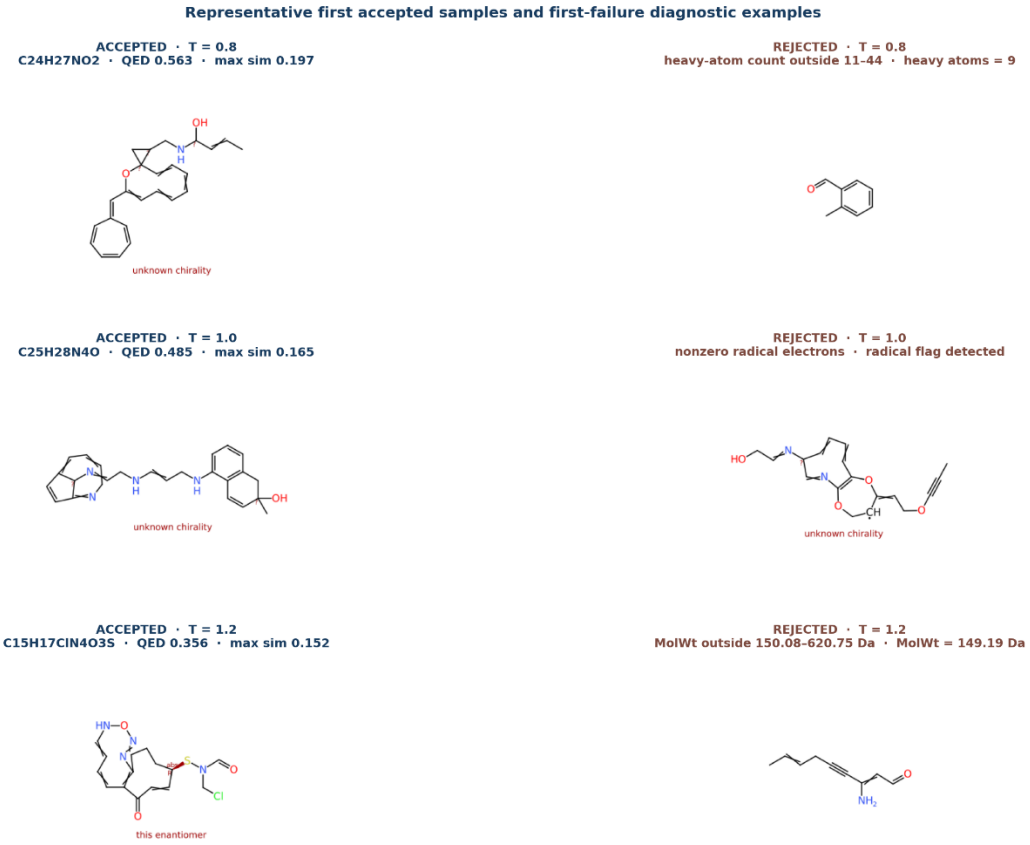
131	66	8	0
HEAVY-ATOM RANGE	RADICALS	MOLWT RANGE	ALL OTHER STAGES



The number of rejected samples increased from 37 at T = 0.8 to 61 at T = 1.0 and 107 at T = 1.2. The rise was driven mainly by radical-containing structures and out-of-range heavy-atom counts, showing that higher sampling temperature preserved RDKit validity while moving more outputs beyond the conservative chemical-distribution envelope.

8.3 Accepted and Rejected Examples

The accepted examples below passed every staged filter and were later confirmed to be exactly and structurally novel. The rejected examples are still RDKit-valid depictions; they fail only at later distribution or radical checks. This distinction demonstrates why 100% RDKit validity does not imply 100% chemistry-aware acceptance.



Structural Novelty and Nearest-Neighbor Analysis

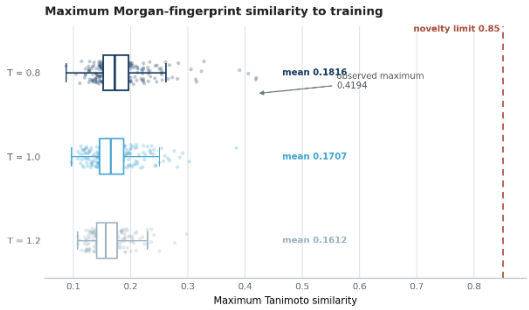
09 / STRUCTURE-SPACE AND EMBEDDING-SPACE COMPARISON

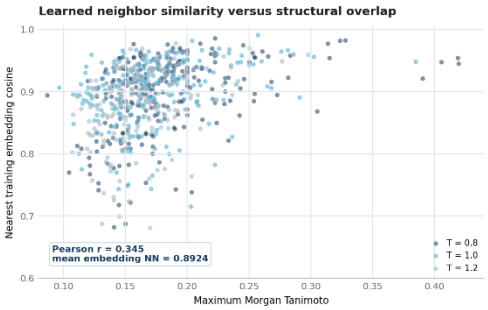
Structural novelty was evaluated for every unique accepted molecule against all 3,355 training structures, while the nearest-neighbor analysis examined where the same molecules fell in the learned joint representation space. Keeping these two tests separate is essential: one measures explicit circular-substructure overlap, whereas the other measures similarity under the GCN-Transformer embedding learned by the model.

695 / 695	0.1722	0.1644	0.4194
STRUCTURALLY NOVEL	MEAN MAX TANIMOTO	MEDIAN MAX TANIMOTO	HIGHEST OBSERVED

9.1 Fingerprint-Based Novelty

Radius-2, 2,048-bit Morgan fingerprints were compared with the Tanimoto coefficient T(A,B) = c / (a + b - c). For each generated molecule, the maximum similarity over the complete training split was retained. A molecule was structurally novel only when its canonical SMILES was absent from training and this maximum was strictly below 0.85. All 695 accepted unique molecules satisfied both conditions.





9.2 Nearest Training Molecules

Nearest training molecules were selected by cosine similarity between L2-normalized combined embeddings, z = normalize(z_graph + z_sequence). Mean nearest-neighbor cosine was 0.8924 (median 0.9060; range 0.6229-0.9915), even though maximum fingerprint similarity remained low. Their Pearson correlation was only r = 0.345, indicating that the learned space captures broader cross-modal regularities rather than simply reproducing Morgan-fingerprint overlap.

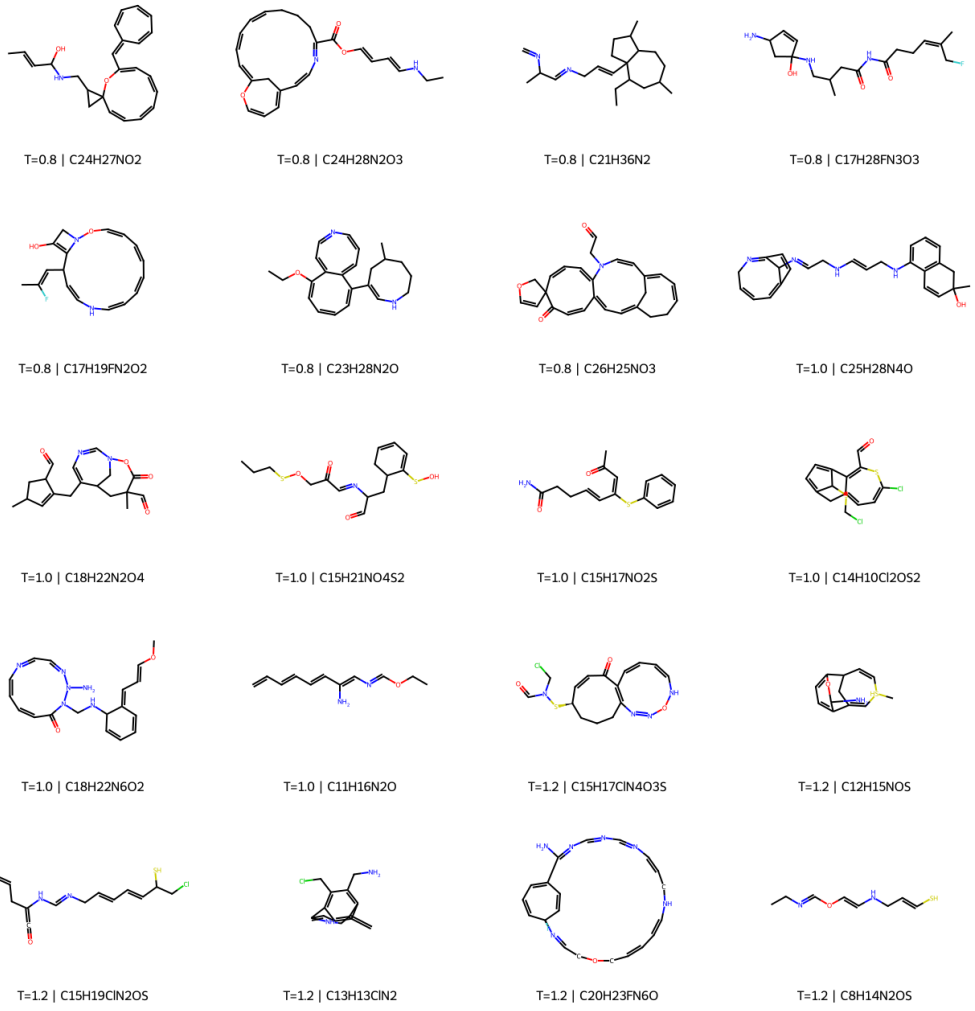
Temp.	Formula	Generated canonical SMILES	Embedding-nearest training SMILES	Cosine / max FP
0.8	C11H19NO4	CC(C)COC(=O)NCCC(C)C(=O)C=O	CCCC(C)(COC(N)=O)COC(=O)NC(C)C	0.9859 0.2222
1.0	C13H11NO	O=Cc1ccccn2ccccc2cc1	O=C(O)c1cc(-c2ccccc2)nc2ccccc12	0.9915 0.2571
1.2	C9H13NO3	CCN1CC(=CC=O)COC1C=O	CCCCC1(CC)C(=O)NC(=O)NC1=O	0.9780 0.1905

Visualization of Generated Molecules

10 / REPRESENTATIVE ACCEPTED AND NOVEL STRUCTURES

The gallery presents 20 representative molecules drawn from the accepted, unique, exactly novel, and structurally novel generation set. The selection spans all three sampling temperatures and preserves the two-dimensional depictions produced from the final canonical structures. Each label reports the sampling temperature and molecular formula, allowing composition and scaffold diversity to be inspected together.

20	7 / 7 / 6	0.4965	0.1675
DISPLAYED MOLECULES	T = 0.8 / 1.0 / 1.2	MEAN QED	MEAN MAX TANIMOTO



The examples include compact heterocycles, fused and bridged ring systems, flexible chains, and structures containing N, O, S, F, or Cl. Their maximum training-set fingerprint similarities range from 0.1111 to 0.3279—well below the 0.85 structural-novelty limit—while QED spans 0.1625 to 0.7995. Thus, the gallery visualizes chemically varied outputs rather than near-copies of training molecules.

Discussion

The results support the central design choice of treating a molecule as two coordinated views rather than forcing graph topology and chemical language through one encoder. The GCN and causal Transformer retained modality-specific computation, while contrastive training made their molecule-level outputs mutually identifiable. At the same time, the generation results show why representation validity, chemical screening, novelty, and practical usefulness must be interpreted as separate claims.

ARCHITECTURAL LOGIC **variable graph -> global readout | causal sequence -> <eos> state | two 192-D projections -> shared cosine space**

11.1 Why Global Pooling Is Required

A molecular graph contains a variable number of atoms, so the GCN naturally returns one hidden vector per node. Contrastive retrieval, however, requires exactly one vector for each molecule, with the same dimensionality as the sequence representation. Global mean pooling supplies this fixed 192-dimensional graph vector and is invariant to atom ordering, which is essential because graph identity must not depend on how nodes are indexed. It also normalizes for graph size, preventing large molecules from dominating merely because they contain more nodes. The trade-off is compression: averaging can weaken rare or localized functional-group signals and can partly obscure molecular size. The present retrieval results show that mean pooling was sufficient for this dataset, but attention pooling, Set2Set, or hierarchical readout could preserve more motif-specific information.

11.2 Limitation of Scalar Bond Weights

The scalar mapping 1.0, 1.5, 2.0, and 3.0 lets single, aromatic, double, and triple bonds change message strength without increasing model complexity. This is a useful baseline, but a scalar is not a complete bond representation. In GCNConv it influences normalized aggregation; it does not provide a learned categorical channel for conjugation, ring membership, stereochemistry, or bond direction. Different chemical situations may therefore receive the same effective treatment, and the numeric ordering implicitly assumes that bond type is adequately represented by magnitude. The strong cross-modal retrieval obtained here shows that this approximation did not prevent useful learning, but it does not establish that the edge encoding is chemically optimal. Edge-conditioned message passing or learned bond embeddings would be a more expressive comparison.

11.3 Importance of Causal Masking

Causal masking is the condition that makes next-token training consistent with generation. At position t, the Transformer may use only the prefix through t; future SELFIES symbols are hidden, while padding is excluded through a separate key-padding mask and ignored in the loss. Without the causal mask, teacher forcing would leak the correct continuation into the hidden states, producing optimistic token metrics that could not be reproduced when sampling from <bos>. The implemented mask therefore does more than regularize attention: it enforces the same information boundary during training and inference. Selecting the <eos> hidden state as the sequence-level vector is also coherent under this mask because that state summarizes the entire preceding molecular string without observing padded positions.

11.4 Autoregressive Generation

Generation was genuinely autoregressive: each SELFIES token was sampled from logits computed from the tokens already generated, then appended to form the next input. The graph encoder, property target, test molecules, and ground-truth continuations were absent. Consequently, the 900 outputs test what the causal language branch learned as an unconditional distribution, not graph-conditioned translation or property optimization. All trajectories emitted <eos> before 100 content tokens and all decoded structures passed RDKit parsing and sanitization. That 100% validity is consistent with the robustness of SELFIES plus effective sequence learning, but the lower 77.22% chemistry-aware acceptance rate demonstrates that grammatical and toolkit validity are necessary rather than sufficient.

GENERATION EVIDENCE **900/900 <eos> termination | 900/900 RDKit-valid | 695/900 chemistry-aware accepted**

11.5 Importance of Language-Model Warm-up

The 15-epoch language-model warm-up gave the sequence branch a stable autoregressive solution before it was asked to satisfy cross-modal discrimination. Validation token loss fell to 1.6363, token accuracy reached

50.98%, and perplexity fell to 5.1363 during this stage. Joint training then preserved the language objective: validation perplexity was 5.1008 at the selected epoch 37, while the matching-nonmatching cosine gap expanded from 0.0327 to 0.8325. This separation of trajectories is important. Most joint-stage progress came from the contrastive term rather than from a deterioration of token modeling. Warm-up therefore acted as an optimization curriculum, reducing the risk that a randomly initialized sequence model would be shaped primarily by alignment before learning valid chemical syntax.

11.6 Effect of Batch Size on Contrastive Negatives

With a batch size of 256, each graph or sequence in a full batch competed against 255 nonmatching examples in addition to its paired positive. This supplies much richer discrimination than pairwise positive training and helps explain the large validation cosine gap and strong exhaustive test retrieval. The test procedure compared all 420 candidates, so the reported performance is not limited to the easier in-batch evaluation setting. Still, in-batch negatives are sampled incidentally rather than selected for chemical difficulty. Some structurally related molecules may behave as false or overly hard negatives, whereas many unrelated molecules contribute little gradient. A memory bank, larger effective batch, or chemistry-aware hard-negative mining could improve Top-1 identity resolution.

11.7 Effect of Sampling Temperature

Sampling temperature produced a clear quality-yield trade-off. At T = 0.8, 263 of 300 outputs were accepted (87.67%) with mean QED 0.5129; at T = 1.2, acceptance fell to 193 of 300 (64.33%) and mean QED to 0.4183. Rejections increasingly reflected out-of-range heavy-atom counts and radical electrons, even though RDKit validity remained 100%. Mean sequence length also decreased from 46.46 to 42.99 tokens, consistent with the observed higher-temperature distribution making early <eos> selection more likely in this run. T = 1.0 provided an intermediate 79.67% acceptance rate. Because uniqueness and novelty were already 100% for all accepted sets, higher temperature did not improve those saturated metrics; it mainly reduced the fraction remaining inside the conservative training-derived chemical envelope. QED was descriptive and must not be read as a selection objective.

11.8 Exact Novelty and Near-Memorization

Exact novelty alone would only show that canonical SMILES strings were absent from training; it would not exclude close analogues or alternative encodings of nearly memorized scaffolds. The post-acceptance fingerprint analysis addressed this stronger concern. Every one of the 695 unique accepted molecules had maximum radius-2 Morgan Tanimoto similarity below 0.85, with mean 0.1722, median 0.1644, and an observed maximum of only 0.4194. These values provide substantial evidence against direct structural copying under the selected fingerprint. Nevertheless, novelty is metric-dependent: a circular fingerprint emphasizes local substructures and does not by itself establish scaffold novelty under every definition, stereochemical novelty, synthetic feasibility, or biological relevance. Reporting both exact and fingerprint-based novelty therefore strengthens the claim without overstating it.

> INTERPRETIVE BOUNDARY **novel does not mean useful | valid does not mean synthesizable | similar in embedding space does not mean structurally copied**

11.9 Compatibility of GCN and Transformer Representations

The learned representations were strongly compatible across modalities. On the fixed test set, Graph-to-SELFIES and SELFIES-to-Graph Top-5 retrieval reached 85.95% and 83.57%, and matching pairs had mean cosine 0.8806 compared with 0.0306 for nonmatching pairs. The 0.8500 gap and small directional difference show that neither encoder alone dominated the shared space. Top-1 accuracy near 50%, however, leaves meaningful room for exact identity separation. The nearest-training analysis adds an important nuance: generated molecules had high learned-space neighbor cosine (mean 0.8924) but low fingerprint similarity, with correlation only r = 0.345. Thus, the common space appears to capture broad molecular regularities shared by graph and sequence views rather than merely reproducing Morgan-fingerprint neighborhoods. Selecting epoch 37 by retrieval, despite a slightly lower total loss at epoch 40, was therefore consistent with the model's intended use.

11.10 Chemical and Practical Limitations

The chemical conclusions remain bounded by the data and evaluation design. The model was trained on 4,194 molecules from one Lipophilicity benchmark, using one fixed random split; generalization to new scaffolds, larger molecules, charged species, or other chemical domains was not established. Generation was unconditional and the retained lipophilicity targets did not guide sampling, so no claim can be made that generated structures optimize lipophilicity or any therapeutic property. The acceptance filters were conservative distribution checks derived from training percentiles, not proofs of valence stability, synthetic accessibility, purity, or laboratory realizability. RDKit validity, QED, uniqueness, novelty, and embedding similarity are complementary computational diagnostics, but none substitutes for expert chemical review, reaction planning, quantum calculations, docking, ADMET assessment, or experimental validation. Finally, the modest dataset size, simplified bond encoding, global mean pooling, fixed novelty threshold, and absence of repeated seeds or external validation limit statistical certainty. The results should therefore be interpreted as a successful proof of concept for multimodal molecular representation learning and controlled candidate generation, not as evidence of ready-to-use compounds.

Limitations and Future Work

12 / EVIDENCE GAPS, NEXT STEPS, AND VALIDATION PRIORITIES

The experiments establish a coherent proof of concept for graph-SELFIES alignment and unconditional molecular generation, but the current evidence does not establish broad chemical generalization, property optimization, or laboratory usability. The next stage should preserve the fixed test protocol while converting each present boundary into a controlled experiment with an explicit success criterion.

PRIORITY ROADMAP **1) repeated-seed scaffold evaluation | 2) edge-aware and ablated encoders | 3) property-conditioned generation | 4) synthesis-aware screening and external validation**

12.1 Dataset Coverage and Generalization

Limitation. Training used 4,194 molecules from a single Lipophilicity benchmark and one fixed random split. A random split can place related scaffolds in different partitions, and the observed performance therefore cannot be assumed to transfer to larger, charged, stereochemically complex, or out-of-domain compounds. **Future work.** Repeat training across several seeds; report mean, standard deviation, and confidence intervals; add scaffold and similarity-aware splits; and evaluate without fine-tuning on at least one external molecular dataset. Success should mean retained retrieval and validity under genuinely novel scaffolds, not only another random split.

12.2 Representation Capacity and Architectural Evidence

Limitation. Scalar bond weights compress bond type into one magnitude, global mean pooling can dilute localized motifs, and a canonical linearization exposes the sequence encoder to only one traversal of each molecule. The present results show that these choices work, but not that they are optimal or chemically complete. **Future work.** Compare learned edge embeddings and edge-aware GNNs (for example GINE or a graph Transformer), attention or Set2Set readout, and randomized sequence augmentation. A formal ablation should change one component at a time and measure token perplexity, bidirectional Top-1/Top-5 retrieval, cosine gap, parameter count, and runtime.

12.3 Alignment, Negatives, and Statistical Reliability

Limitation. Batch-wise contrastive learning supplied many negatives, yet they were not selected for chemical difficulty, and exact Top-1 retrieval remained 212/420 for Graph-to-SELFIES and 205/420 in the reverse direction. Model selection and conclusions were also based on one training trajectory. **Future work.** Use chemistry-aware hard-negative mining, a cross-batch queue or memory bank, and controlled temperature/margin studies. Compare against unaligned, graph-only, sequence-only, and alternative pooling baselines across repeated seeds. Improvements should be accepted only when Top-1 rises without degrading perplexity, Top-5 retrieval, or the matching/nonmatching cosine separation.

12.4 Conditional and Multi-objective Molecular Generation

Limitation. The 900 samples were generated unconditionally. Neither a source graph nor the retained lipophilicity value controlled decoding, and temperature/top-k/top-p altered diversity without optimizing a specified chemical objective. QED was descriptive rather than a training or acceptance target. **Future work.** Condition the decoder on a target lipophilicity interval, a graph embedding, or both, then measure target error together with validity, acceptance,

uniqueness, novelty, and diversity. A later multi-objective stage can combine constrained decoding with Bayesian optimization or reinforcement learning, while explicitly balancing QED, synthetic accessibility, ADMET risk, and distance from the training set.

12.5 Chemical Validation and Experimental Translation

Limitation. SELFIES termination, RDKit sanitization, percentile filters, QED, and radius-2 Morgan Tanimoto are computational checks; they do not prove tautomeric or stereochemical correctness, scaffold-level novelty, synthetic accessibility, stability, biological activity, or safety. The structural-novelty threshold of 0.85 is also fingerprint-dependent. **Future work.** Extend validation with charge/tautomer/stereochemistry standardization, Bemis-Murcko scaffold analysis, internal diversity, SA or SCScore, PAINS/Brenk alerts, and retrosynthetic route assessment. Prioritize candidates by expert review and property-specific docking, ADMET, molecular dynamics, or quantum calculations, then validate a small preregistered shortlist experimentally. Laboratory evidence should remain the final criterion.

Conclusion

13 / FINAL SYNTHESIS AND PROJECT OUTCOME

This project developed and evaluated a unified GCN-Transformer framework that learns the same molecular identity from two complementary views: a topology-preserving graph and an autoregressive SELFIES sequence. The two-stage curriculum first established a stable chemical language model and then aligned the graph and sequence encoders through symmetric contrastive learning while retaining next-token prediction.

CORE OUTCOME **one shared molecular space | strong bidirectional retrieval | valid autoregressive generation | accepted structures novel to training**

REPRESENTATION LEARNING The selected epoch-37 checkpoint preserved validation perplexity at 5.1008 while producing a matching-pair cosine of 0.8776 and a matching-nonmatching gap of 0.8325. On the fixed 420-molecule test set, Graph-to-SELFIES retrieval reached 50.48% Top-1 and 85.95% Top-5, and SELFIES-to-Graph reached 48.81% and 83.57%. Matching pairs averaged 0.8806 cosine similarity versus 0.0306 for nonmatches. These consistent results in both directions show that the learned space captures cross-modal molecular identity rather than favoring only one representation.

GENERATIVE PERFORMANCE Unconditional autoregressive sampling produced 900 terminated SELFIES sequences across temperatures 0.8, 1.0, and 1.2, all of which decoded to RDKit-valid molecules. The staged validation pipeline retained 695 structures, giving a 77.22% overall acceptance rate. Every accepted canonical molecule was unique, absent from the training set, and below the 0.85 Morgan-fingerprint similarity threshold; the highest observed maximum Tanimoto was only 0.4194. Acceptance and mean QED decreased as temperature increased, demonstrating the expected trade-off between conservative sampling and chemical-distribution compliance.

INTERPRETATION Together, the retrieval and generation experiments support the central hypothesis that a residual GCN and a causal Transformer can be brought into a compatible molecular space without discarding the Transformer's generative function. At the same time, the validation results define the correct boundary of this conclusion: SELFIES validity, RDKit sanitization, novelty, QED, and fingerprint distance are computational evidence, not proof of stability, synthetic accessibility, biological activity, safety, or practical value.

Final conclusion. The project met its central computational objective: it learned compatible graph and SELFIES representations without losing autoregressive sequence modeling, and it used that model to generate a substantial set of chemically parseable, filtered, and training-novel molecules. The appropriate next step is not to treat these outputs as drug candidates, but to test whether the same representation remains reliable under scaffold-aware, external, synthesis-aware, and experimental evaluation.